Linear Algebra and Its Applications

Section 5.4: Eigenvectors and linear transformations

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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5.4 Eigenvectors and linear transformations

The matrix of a linear map depends on its *ordered* domain and codomain bases. Column j is the coordinate vector of the image of domain basis vector j, written in the codomain basis. For $T(x) = Ax$ on $\mathbb{R}^n$, if P has the basis vectors as its columns, then $[T]_{\mathcal{B}} = P^{-1}AP$.

The relation $[T(x)]_{\mathcal{B}} = [T]_{\mathcal{B}}[x]_{\mathcal{B}}$ uses coordinates on both sides. A coordinate column is not automatically the same as the vector it represents; they coincide only when using the standard basis. Similarity changes coordinates without changing the underlying linear transformation.

Practice problem 5.4.P1: Recover a polynomial from its coordinate column

Relative to $(1, t, t^2)$, the polynomial $p = a_0 + a_1t + a_2t^2$ has column $(a_0, a_1, a_2)^T$. Multiply by the supplied matrix:

$$[T(p)]_{\mathcal{B}} = \begin{bmatrix} 3 & 4 & 0 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 3a_0 + 4a_1 \\ 5a_1 - a_2 \\ a_0 - 2a_1 + 7a_2 \end{bmatrix}.$$

Translate this column back into the polynomial $T(p) = (3a_0 + 4a_1) + (5a_1 - a_2)t + (a_0 - 2a_1 + 7a_2)t^2$.

Practice problem 5.4.P2: Reflexivity and transitivity of similarity

(a) Every matrix is similar to itself.

Choose $P = I$; then $P^{-1}AP = A$ and P is invertible.

(b) Compose two changes of basis.

If $B = P^{-1}AP$ and $C = Q^{-1}BQ$, substitute the first equation into the second:

$$C = Q^{-1}P^{-1}APQ = (PQ)^{-1}A(PQ).$$

The product PQ is invertible, so this is exactly the definition saying C is similar to A. The reversed order in $(PQ)^{-1} = Q^{-1}P^{-1}$ is essential.

Exercise 5.4.1: Construct a matrix from basis-vector images

The codomain coordinate vectors of $T(b_1)$, $T(b_2)$, $T(b_3)$ are $(3, -5)^T$, $(-1, 6)^T$, and $(0, 4)^T$. Put them in columns, in that order:

$$[T]_{\mathcal{D}\leftarrow\mathcal{B}} = \begin{bmatrix} 3 & -1 & 0 \\ -5 & 6 & 4 \end{bmatrix}.$$

It has two rows (the codomain dimension) and three columns (the domain dimension). Multiplication by a column $(x_1, x_2, x_3)^T$ produces the coordinates of $x_1T(b_1) + x_2T(b_2) + x_3T(b_3)$, explaining the column rule.

Exercise 5.4.2: Construct a matrix from basis-vector images

The domain basis is now $\mathcal{D}$ and the codomain basis is $\mathcal{B}$. The supplied images have coordinate columns $(3,-3)^T$ and $(-2,5)^T$ in $\mathcal{B}$. Therefore

$$[T]_{\mathcal{B}\leftarrow\mathcal{D}} = \begin{bmatrix} 3 & -2 \\ -3 & 5 \end{bmatrix}.$$

The arrow in the subscript records the direction of coordinates and helps avoid reversing the bases.

Exercise 5.4.3: Construct a matrix from basis-vector images

(a) Substitute the standard basis vectors one at a time.

The formula gives $T(e_1) = b_3$, $T(e_2) = -b_1 - 2b_2$, and $T(e_3) = 2b_1 + 3b_3$.

(b) Read their coefficients in $\mathcal{B}$.

These columns are $(0,0,1)^T$, $(-1,-2,0)^T$, and $(2,0,3)^T$.

(c) Put them in the original basis order.

Thus the matrix relative to the standard domain basis and codomain basis $\mathcal{B}$ is

$$\begin{bmatrix} 0 & -1 & 2 \\ 0 & -2 & 0 \\ 1 & 0 & 3 \end{bmatrix}.$$

Multiplying by $(x_1,x_2,x_3)^T$ recovers exactly $(2x_3 - x_2, -2x_2, x_1 + 3x_3)^T$, the coefficients in the given transformation.

Exercise 5.4.4: Construct a matrix from basis-vector images

The input column in $\mathcal{B}$ is $(x_1,x_2,x_3)^T$, while the output is already in standard coordinates of $\mathbb{R}^2$. Reading the coefficient of each input scalar gives

$$[T]_{\mathcal{E}\leftarrow\mathcal{B}} = \begin{bmatrix} 2 & -3 & 1 \\ -2 & 0 & 5 \end{bmatrix}.$$

In particular the zero in row 2, column 2 is needed because the second output component does not contain x_2.

Exercise 5.4.5: Multiplication of polynomials as a linear map

(a) Multiply the specified polynomial, combining like powers.

$$T(3 - 2t + t^2) = (t+3)(3 - 2t + t^2) = t^3 + t^2 - 3t + 9.$$

(b) Verify linearity for arbitrary inputs and scalars.

For polynomials p, q and scalars c, d, distributivity gives

$$T(cp + dq) = (t + 3)(cp + dq) = cT(p) + dT(q).$$

The degree is at most 3, so the image does belong to the stated codomain.

(c) Apply the map to each monomial in the domain basis.

$T(1) = t + 3$. $T(t) = t^2 + 3t$. $T(t^2) = t^3 + 3t^2$. Writing those coefficients in increasing powers of t gives the matrix

$$C = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

The column lengths reflect the codomain degree bound, even when one image has lower degree and hence trailing zero coefficients.

Exercise 5.4.6: Multiplication of polynomials as a linear map

(a) Multiply the specified polynomial, combining like powers.

$$T(3 - 2t + t^2) = (2t^2 + 1)(3 - 2t + t^2) = 2t^4 - 4t^3 + 7t^2 - 2t + 3.$$

(b) Verify linearity for arbitrary inputs and scalars.

For polynomials p, q and scalars c, d, distributivity gives

$$T(cp + dq) = (2t^2 + 1)(cp + dq) = cT(p) + dT(q).$$

The degree is at most 4, so the image does belong to the stated codomain.

(c) Apply the map to each monomial in the domain basis.

$T(1) = 2t^2 + 1$. $T(t) = 2t^3 + t$. $T(t^2) = 2t^4 + t^2$. Writing those coefficients in increasing powers of t gives the matrix

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

The column lengths reflect the codomain degree bound, even when one image has lower degree and hence trailing zero coefficients.

Exercise 5.4.7: Coordinates determine a linear transformation

Evaluate the formula at the three basis polynomials: $T(1) = 3 + 5t$, $T(t) = -2t + 4t^2$, $T(t^2) = t^2$. Their coefficient columns give

$$[T]_{\mathcal{B}} = \begin{bmatrix} 3 & 0 & 0 \\ 5 & -2 & 0 \\ 0 & 4 & 1 \end{bmatrix}.$$

As a check, multiplying this by $(a_0, a_1, a_2)^T$ yields $(3a_0, 5a_0 - 2a_1, 4a_1 + a_2)^T$, exactly the supplied coefficient rule.

Exercise 5.4.8: Coordinates determine a linear transformation

The input has coordinate vector $(4, -3, 0)^T$. Thus

$$[T(4b_1 - 3b_2)]_{\mathcal{B}} = \begin{bmatrix} 0 & 0 & 1 \\ 2 & 1 & -2 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ -5 \end{bmatrix}.$$

Translate back to the vector space: $T(4b_1 - 3b_2) = 5b_2 - 5b_3$. The result is not the coordinate column itself unless these basis vectors happen to be standard coordinate vectors.

Exercise 5.4.9: Evaluation at three input values

(a) Evaluate the given polynomial.

For $p(t) = 5 + 3t$, the three values are $p(-1) = 2$, $p(0) = 5$, and $p(1) = 8$, so $T(p) = (2, 5, 8)^T$.

(b) Establish linearity.

For any fixed input s, $(cp + dq)(s) = cp(s) + dq(s)$. Applying this at $s = -1, 0, 1$ componentwise proves $T(cp + dq) = cT(p) + dT(q)$.

(c) Evaluate the three basis polynomials.

The image columns for $1, t, t^2$ are $(1, 1, 1)^T$, $(-1, 0, 1)^T$, and $(1, 0, 1)^T$. Therefore

$$[T] = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

For example this matrix times $(5, 3, 0)^T$ returns the value found in part (a).

Exercise 5.4.10: Evaluation of a cubic at four points

(a) Linearity holds componentwise.

Evaluation at any fixed input distributes over polynomial addition and scalar multiplication. Applying this at $-2, 3, 1, 0$ proves the vector-valued map is linear.

(b) Respect the given order of evaluation points.

Evaluate each of $1, t, t^2, t^3$, putting its four values in a column. The matrix is

$$[T] = \begin{bmatrix} 1 & -2 & 4 & -8 \\ 1 & 3 & 9 & 27 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

For a general coefficient column $(a_0, a_1, a_2, a_3)^T$, multiplication gives the polynomial's values at those four points. The row for input 0 is last, not first, because the output order was specified that way.

Exercise 5.4.11: Change both input and output coordinates

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 2 & 3 \\ -4 & -5 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -2 & -2 \\ 0 & -1 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order.

Exercise 5.4.12: Change both input and output coordinates

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} -2 & 2 \\ 0 & -4 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -4 & 0 \\ 2 & -2 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order.

Exercise 5.4.13: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 0 & 1 \\ -3 & 4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-3)(t-1).$$

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -\frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

The exact checks are $\det P = 2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.14: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 5)(t + 1).$$

Find the eigenspace for $\lambda = -1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-1)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-1} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{5} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 \\ 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 2 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.15: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-2)(t+5).$$

Find the eigenspace for $\lambda = -5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-5)I) = \begin{bmatrix} 1 & \frac{1}{3} \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-5} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -3 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (-5)v$ for each listed vector.

Find the eigenspace for $\lambda = 2$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (2)I) = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{2} = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (2)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 2 \\ -3 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -5 & 0 \\ 0 & 2 \end{bmatrix}.$$

The exact checks are $\det P = 7 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.16: Choose an eigenvector basis

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} 4 & -2 \\ -1 & 5 \end{bmatrix}.$$

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t - 6)(t - 3).$$

Find the eigenspace for $\lambda = 3$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (3)I) = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_3 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (3)v$ for each listed vector.

Find the eigenspace for $\lambda = 6$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (6)I) = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Columns 2 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_6 = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (6)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $1 + 1 = 2$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \qquad D = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}.$$

The exact checks are $\det P = -3 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. The columns of P are the requested ordered basis $\mathcal{B}$, and its basis matrix is $[T]_{\mathcal{B}} = D$.

Exercise 5.4.17: An eigenvector without an eigenvector basis

(a) Verify the stated eigenvector and inspect multiplicity.

Multiplication gives $A(1,-1)^T = (3,-3)^T = 3(1,-1)^T$. But

$$\det(tI - A) = (t-3)^2, \qquad A - 3I = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}.$$

The homogeneous equation is $x_1 + x_2 = 0$, so its eigenspace has only dimension 1. It cannot supply a basis for $\mathbb{R}^2$; thus A is not diagonalizable.

(b) Use the specified basis anyway.

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \qquad \det P = 1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 3 & -2 \\ -3 & 5 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. The resulting matrix is not diagonal. A valid change of basis exists even when no diagonalizing change of basis exists.

Exercise 5.4.18: Similarity and coordinate identities

The eigenvalues alone do not decide the question. A diagonal basis matrix exists precisely when A is diagonalizable, which here means the eigenspace for 5 has dimension 2. The eigenspace for -2, whose algebraic multiplicity is 1, has dimension 1.

For a positive example take $A = \operatorname{diag}(5,5,-2)$; the standard basis works. For a negative example take $A = \left[\begin{smallmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -2 \end{smallmatrix}\right]$. It has the same eigenvalues, but its 5-eigenspace is only one-dimensional. Thus one must compute an eigenspace or equivalent rank information before answering for a particular A.

Exercise 5.4.19: Similarity and coordinate identities

Write $B = P^{-1}AP$ with P invertible. The product is invertible because all three factors are, and reversing the factors gives

$$B^{-1} = P^{-1}A^{-1}P.$$

Thus the same change of basis $Q = P$ exhibits similarity of the inverses. It is legitimate to invert this expression because invertibility was established first.

Exercise 5.4.20: Similarity and coordinate identities

If $B = P^{-1}AP$, then $B^2 = P^{-1}AP\,P^{-1}AP = P^{-1}A^2P$, since the middle $PP^{-1} = I$. This is the required similarity factorization for the squares. The same cancellation would prove the result for any positive integer power.

Exercise 5.4.21: Similarity and coordinate identities

Write $B = P^{-1}AP$ and $C = Q^{-1}AQ$. Solving the second identity for A gives $A = QCQ^{-1}$. Substitution yields

$$B = P^{-1}QCQ^{-1}P = R^{-1}CR, \qquad R = Q^{-1}P.$$

The matrix R is invertible as a product of invertible matrices, so B and C are similar. This argument is the symmetry-plus-transitivity property in explicit form.

Exercise 5.4.22: Similarity and coordinate identities

Suppose $A = PDP^{-1}$ with D diagonal and $B = Q^{-1}AQ$. Then

$$B = (Q^{-1}P)D(P^{-1}Q).$$

The last factor is $(Q^{-1}P)^{-1}$. Thus this is a diagonalization of B, with an invertible change-of-basis matrix $Q^{-1}P$.

Exercise 5.4.23: Similarity and coordinate identities

Set $y = P^{-1}x$. This vector is nonzero: if $y = 0$, multiplication by P would force $x = 0$, contradicting the eigenvector definition. Now

$$By = P^{-1}AP\,P^{-1}x = P^{-1}Ax = P^{-1}(\lambda x) = \lambda y.$$

Therefore y is an eigenvector of B for the same eigenvalue. The nonzero check is an essential part of the proof.

Exercise 5.4.24: Similarity and coordinate identities

For $B = P^{-1}AP$, first note that AP has the same column space as A. Every column of AP is a combination of columns of A, and $A = (AP)P^{-1}$ supplies the reverse inclusion. Thus $\operatorname{rank}(AP) = \operatorname{rank} A$. Next, multiplication on the left by P^{-1} is an invertible linear map on the codomain. It maps the column space of AP bijectively onto that of B, preserving its dimension. Consequently $\operatorname{rank} B = \operatorname{rank} A$.

Exercise 5.4.25: Similarity and coordinate identities

Use the supplied cyclic identity on two square factors, $F = P^{-1}A$ and $G = P$:

$$\operatorname{tr}(B) = \operatorname{tr}((P^{-1}A)P) = \operatorname{tr}(P(P^{-1}A)) = \operatorname{tr}(A).$$

Only associativity and $PP^{-1} = I$ are used; this does not incorrectly commute arbitrary matrix factors.

Exercise 5.4.26: Similarity and coordinate identities

If $A = PDP^{-1}$ with $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, Exercise 25 gives $\operatorname{tr} A = \operatorname{tr} D$. The latter is the sum of the diagonal entries, hence $\sum_{j=1}^{n} \lambda_j$. These diagonal entries are the eigenvalues counted with their algebraic multiplicities in a diagonalization, proving the assertion in the requested case.

Exercise 5.4.27: Similarity and coordinate identities

The identity map sends the jth domain basis vector b_j to the same vector b_j, now expressed in the standard codomain basis. Thus its columns are $b_1, \dots, b_n$:

$$[I]_{\mathcal{E} \leftarrow \mathcal{B}} = [b_1 \ \cdots \ b_n] = P_{\mathcal{B}}.$$

This is the change-of-coordinates matrix from $\mathcal{B}$ coordinates to standard coordinates, also called the basis matrix in Section 4.4. Its action is $x = P_{\mathcal{B}}[x]_{\mathcal{B}}$.

Exercise 5.4.28: Similarity and coordinate identities

The identity sends b_j to b_j, whose codomain coordinate column is $[b_j]_{\mathcal{C}}$. Therefore

$$[I]_{\mathcal{C} \leftarrow \mathcal{B}} = \big[[b_1]_{\mathcal{C}} \ \cdots \ [b_n]_{\mathcal{C}}\big].$$

This is the change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$ in Section 4.7. Its defining equation is $[x]_{\mathcal{C}} = P_{\mathcal{C} \leftarrow \mathcal{B}}[x]_{\mathcal{B}}$; the vector x itself does not change.

Exercise 5.4.29: Similarity and coordinate identities

With the same basis on both sides, $[I(b_j)]_{\mathcal{B}} = [b_j]_{\mathcal{B}} = e_j$. The matrix with those columns is exactly I_n. Hence $[I]_{\mathcal{B}} = I_n$ for every ordered basis, not merely the standard one.

Exercise 5.4.30: An exact basis-matrix computation

The standard matrix is

$$A = \begin{bmatrix} 6 & -2 & -2 \\ 3 & 1 & -2 \\ 2 & -2 & 2 \end{bmatrix}.$$

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 1 & -1 \\ 1 & 3 & 0 \end{bmatrix}, \qquad \det P = -1 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 2 & 4 & -4 \\ 2 & 1 & -4 \\ 2 & 8 & 0 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3 & 0 \\ 0 & 1 & 4 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. This computer-assisted calculation can be reproduced exactly by setting the displayed matrices in SymPy and evaluating

```
C = P.inv()*A*P
assert P*C == A*P
print(C)
```

Solving the augmented system instead of explicitly forming the inverse gives the same result.

Exercise 5.4.31: An exact basis-matrix computation

The standard matrix is

$$A = \begin{bmatrix} -7 & -48 & -16 \\ 1 & 14 & 6 \\ -3 & -45 & -19 \end{bmatrix}.$$

Place the ordered basis vectors in columns:

$$P = \begin{bmatrix} -3 & -2 & 3 \\ 1 & 1 & -1 \\ -3 & -3 & 0 \end{bmatrix}, \qquad \det P = 3 \neq 0.$$

The columns of AP are the images of those basis vectors in standard coordinates:

$$AP = \begin{bmatrix} 21 & 14 & 27 \\ -7 & -6 & -11 \\ 21 & 18 & 36 \end{bmatrix}.$$

To express those images in the given basis, solve $PC = AP$ column by column. Equivalently row reduce $[P \mid AP]$ to $[I \mid C]$. The result is

$$C = P^{-1}AP = \begin{bmatrix} -7 & -2 & -6 \\ 0 & -4 & -6 \\ 0 & 0 & -1 \end{bmatrix}.$$

This is the required basis matrix. Exact multiplication verifies $PC = AP$, so each of its columns really is the coordinate vector of the corresponding image, in the stated basis order. This computer-assisted calculation can be reproduced exactly by setting the displayed matrices in SymPy and evaluating

```
C = P.inv()*A*P
assert P*C == A*P
print(C)
```

Solving the augmented system instead of explicitly forming the inverse gives the same result.

Exercise 5.4.32: Find a diagonal coordinate representation

What we must show. Determine whether A has a basis of eigenvectors, where

$$A = \begin{bmatrix} -6 & 4 & 0 & 9 \\ -3 & 0 & 1 & 6 \\ -1 & -2 & 1 & 0 \\ -4 & 4 & 0 & 7 \end{bmatrix}.$$

This is a matrix-program exercise; the eigenvalue and null-space calculations below were carried out with exact rational arithmetic, then checked by multiplication.

Find eigenvalues with multiplicities.

Use $p(t) = \det(tI - A)$; changing to $\det(A - tI)$ changes only an overall sign and not its roots. Here

$$p(t) = (t-5)(t-1)(t+2)^2.$$

Find the eigenspace for $\lambda = -2$.

The algebraic multiplicity is 2. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (-2)I) = \begin{bmatrix} 1 & 0 & -1 & -\frac{3}{2} \\ 0 & 1 & -1 & \frac{3}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 3, 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_{-2} = \operatorname{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -3 \\ 0 \\ 4 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 2. Direct multiplication checks $Av = (-2)v$ for each listed vector.

Find the eigenspace for $\lambda = 1$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (1)I) = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{7}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_1 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ -1 \\ -7 \\ 2 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (1)v$ for each listed vector.

Find the eigenspace for $\lambda = 5$.

The algebraic multiplicity is 1. Solve $(A - \lambda I)x = 0$; row reduction gives

$$\operatorname{rref}(A - (5)I) = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 4 are nonpivot columns, so their variables can be chosen freely. Solving the pivot equations gives

$$E_5 = \operatorname{Span}\left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \\ 2 \end{bmatrix} \right\}.$$

The displayed vectors are independent (their free coordinates give independent parameter choices), and every solution is their linear combination. Thus this eigenspace has dimension 1. Direct multiplication checks $Av = (5)v$ for each listed vector.

Count independent eigenvectors.

Bases for eigenspaces at different eigenvalues combine into an independent set. The total dimension is $2 + 1 + 1 = 4$. This equals the matrix size, so it provides the required basis. In the order just listed, take

$$P = \begin{bmatrix} 1 & 6 & 2 & 2 \\ 1 & -3 & -1 & 1 \\ 1 & 0 & -7 & -1 \\ 0 & 4 & 2 & 2 \end{bmatrix}, \qquad D = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}.$$

The exact checks are $\det P = 84 \neq 0$ and $AP = PD$. Multiplying the latter by P^{-1} on the right proves $A = PDP^{-1}$. The order of entries in D must match the order of eigenvectors in P. A reproducible exact command pattern is:

```
for lam,multiplicity,basis in A.eigenvects():
    print(lam,multiplicity,(A-lam*eye(A.rows)).rref())
    print(basis)
```

Here `A` is the displayed matrix and `Matrix, eye` are imported from `sympy`. The printed bases may be rescaled without changing their spans.The columns of the displayed P form the requested basis; the corresponding matrix of the transformation is D.